Lincoln Technical High School · Automotive Technology I (AUTO 110) · Instructor: Mr. Daniels

Assignment 3: Brake Systems

Due: Friday, October 17. **Format:** type your answers in Word, Arial 11. Put your name and class period at the top. Answer in complete sentences unless the question says otherwise.

1. What is the job of a vehicle's brake system? Explain in your own words. (2 pts)

2. Explain the difference between disc brakes and drum brakes. Give one advantage of each. (4 pts)

3. Why do most manufacturers recommend replacing brake fluid every 2 to 3 years? (2 pts)

4. Multiple choice: which brake fluid must NOT be mixed with DOT 4? (1 pt)
a) DOT 3 b) DOT 5 c) DOT 5.1 d) none, they can all be mixed

5. Calculation (4 pts). A force of 150 lb acts on a master cylinder piston with a diameter of 1.0 in.
a) Calculate the hydraulic pressure in the system in psi.
b) The caliper piston has a diameter of 2.5 in. Calculate the clamping force on the brake pad.

6. Identify parts 1 through 6 in Figure 1 (next page). (3 pts)

7. Complete the table. (4 pts)

Symptom	Possible cause
Brake pedal sinks to the floor	
Vehicle pulls to one side when braking	
Squealing noise when braking	
Steering wheel shakes when braking	

8. What does ABS do, and how can a driver tell when ABS is working? (2 pts)

9. Describe step by step how to replace the front brake pads on a car. Include safety. (5 pts)

Total: 27 points

Figure 1: Disc brake (simplified cross-section)

